

Direct numerical simulation of 't Hooft partition function and (de)confining phase

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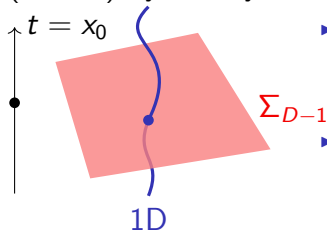
- M. Abe, OM and H. Suzuki, PTEP **2025**, no.6, 063B03 (2025) [arXiv:2501.00286].
- OM and H. Suzuki, PTEP **2025**, no.6, 063B04 (2025) [arXiv:2501.07042].
- <https://github.com/o-morikawa/Gaugefields.jl>

Introduction

- **Symmetry**: fundamental tool in physics
 - ▶ Universal applications to high energy physics, condensed matter physics and mathematics
- More aspects of Symmetry
 - ▶ Landau theory: phase transition (or vacuum structure)
 - ▶ Symmetry breaking: quantum anomaly, spontaneous breaking
 - ▶ 't Hooft anomaly matching for strongly coupled theories ['79]
 - ★ Restriction on low-energy dynamics: SSB, phase structure, SPT
- **Generalized global symmetry** [Gaiotto–Kapustin–Seiberg–Willet '14]
 - ▶ Coupled with topological field theory
 - ★ Changing topological structure without changing local dynamics [Kapustin–Seiberg '14]:
 - ★ Nontrivial information by using generalized 't Hooft anomaly matching [E.g., OM–Wada–Yamaguchi '23]

Generalization: higher-form symmetry

- (0-form) Symmetry



- ▶ Charge (codim 1)

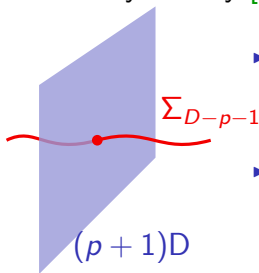
$$Q \equiv \int_{\Sigma_{D-1}} j_0 dx_1 \wedge \cdots \wedge dx_{D-1}$$

- ▶ Symmetry operator

$$U_\alpha(\Sigma_{D-1}) = e^{i\alpha Q}$$

Topological under deformation of Σ_{D-1}

- Higher-form symmetry [Gaiotto–Kapustin–Seiberg–Willet '14]



- ▶ p -form symmetry $G^{[p]}$ (codim $p + 1$)

$$Q \equiv \int_{\Sigma_{D-p-1}} \star j^{(p+1)}, \quad U_\alpha(\Sigma) = e^{i\alpha Q}$$

- ▶ Transforming a “loop” operator $W(C)$

$$W(C) \mapsto U(\Sigma)W(C)$$

$$= e^{i\alpha \#(\Sigma, C)} W(C) \quad \text{w/ linking } \#$$

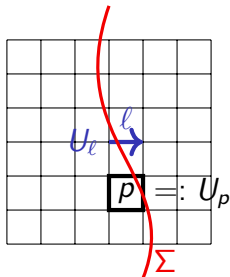
Center symmetry in YM theory

- Lattice $SU(N)$ YM theory
 - link variable $U_\ell \in SU(N)$

- Center symmetry: $\mathbb{Z}_N^{[1]}$

$$e^{\frac{2\pi i}{N}k} \in \mathbb{Z}_N \subset SU(N); \quad U_\ell \mapsto e^{\frac{2\pi i}{N}k \#(\Sigma, \ell)} U_\ell$$

Intersection $\#$ of Σ & link ℓ ; $U_p \mapsto U_p$



- Gauging the center symmetry

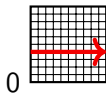
$$S \sim \sum \text{Tr} e^{-\frac{2\pi i}{N} B_p} U_p \quad B_p: \text{2-form gauge field assoc. } \mathbb{Z}_N^{[1]}$$

invariant under $U_\ell \mapsto e^{\frac{2\pi i}{N}\lambda_\ell} U_\ell$, $B_p \mapsto B_p + (d\lambda)_p$

global desc.
 $\Rightarrow$

't Hooft twisted boundary condition [79]:

$$U_{n+L\hat{\nu}, \mu} = g_{n, \nu}^{-1} U_{n, \mu} g_{n+L\hat{\mu}, \nu}$$



gauge transf

$$g_{n+L\hat{\nu}, \mu}^{-1} g_{n, \nu}^{-1} g_{n, \mu} g_{n+L\hat{\mu}, \nu} = e^{\frac{2\pi i}{N} z_{\mu\nu}} \in \mathbb{Z}_N$$

$$\text{'t Hooft flux } z_{\mu\nu} = \sum B_p \bmod N$$

Mixed 't Hooft anomaly due to fractional Q

- Basic property: fractionality of topological charge
 - ▶ Usually, topo. charge $Q \sim \int F\tilde{F} \in \mathbb{Z}$ under topo. sectors
 - ★ Lattice proof: [Lüscher '84]
 - ▶ With background gauge field B , $Q \sim \int \frac{1}{N} B\tilde{B} \in \frac{1}{N}\mathbb{Z}$,

$$\mathcal{Z}_{\theta+2\pi}[B] = e^{-2\pi i Q} \mathcal{Z}_{\theta}[B]$$

[Kapustin–Thorngren '13, Gaiotto–Kapustin–Komargodski–Seiberg '17]

- ★ Cf. Multi branch structure by Witten
- ★ Monopole condensation $\leftrightarrow$ dyon condensation
- ★ Lattice proof:

[Abe–OM–Suzuki '22, Abe–OM–Onoda–Suzuki–Tanizaki '23]

- Under 't Hooft twisted b.c. $Q \sim \int F\tilde{F} = -\frac{\epsilon_{\mu\nu\rho\sigma} Z_{\mu\nu} Z_{\rho\sigma}}{8N}$
[van Baal '82] cf. [Edwards–Heller–Narayanan, de Forcrand–Jahn,
Fodor–Holland–Kuti–Nógrádi–Schroeder, Kitano–Suyama–Yamada, Itou, ...]

Direct Monte Carlo computation of $\mathcal{Z}_{\text{tH}}$

- Traditionally, ratio $\mathcal{Z}[B]/\mathcal{Z}[0]$ (free energy) is computed by “reweighting” [Kovács–Tomboulis '00, de Forcrand–von Smekal '01]
- The 't Hooft partition function [’79]

$$\mathcal{Z}_{\text{tH}}[E_i; B_{ij}] = \frac{1}{N^3} \sum_{B_{i4}=0}^{N-1} \exp \left(\frac{2\pi i}{N} \sum_{i=1}^3 E_i B_{i4} \right) \mathcal{Z}[B]$$

detects quantum phases, such as confinement, Higgs, Coulomb [Tomboulis–Yaffe '85, Kitano–Suyama–Yamada '17, Nguyen–Tanizaki–Ünsal '23]

- We propose that a developed HMC with **dynamical** B -fields which provides $\mathcal{Z}_{\text{tH}}$ directly
 - ▶ All combinations of $B_{\mu\nu}$ for $\mathcal{Z}[B]$ or $\mathcal{Z}_{\text{tH}}[\mathbf{E}; \mathbf{B}]$ automatically.
- A technique to observe (de)confinement phase structure

“Halfway-updating” HMC algorithm

- Let U and B the initial configuration
 - ① Generate π (conjugate to U) by $P_G(\pi) \sim e^{-\pi^2/2}$.
 - ② HMC with $H(U, \pi, B) = \frac{1}{2}\pi^2 + S[U, B]$ by MD time $\tau/2$;
 $\{U, \pi\} \xrightarrow{\tau/2} \{\check{U}, \check{\pi}\}$.
 - ③ Update $B \rightarrow B'$ by $P_F(B \rightarrow B')$, where
 $P_F(B \rightarrow B') = P_F(B' \rightarrow B)$ for $\forall B, B'$.
 - ④ HMC with $H(\check{U}, \check{\pi}, B') = \frac{1}{2}\check{\pi}^2 + S[\check{U}, B']$ by MD time $\tau/2$;
 $\{\check{U}, \check{\pi}\} \xrightarrow{\tau/2} \{U', \pi'\}$.
 - ⑤ Metropolis test $P_A(\{U, \pi, B\} \rightarrow \{U', \pi', B'\}) = \min[1, e^{-\Delta H}]$,
where $\Delta H \equiv H(U', \pi', B') - H(U, \pi, B)$.
 - ⑥ Go back to (1).
- Detailed balance thanks to assumption of P_F
 - ▶ If uniformly random, $SU(N)/\mathbb{Z}_N$ with short autocorrelation (less freezing due to mixing b.c.) [Abe-OM-Suzuki '25].

Counting number of configurations for each flux

- Now, we can have a set of configurations U associated to B
- “Partition function” is defined by

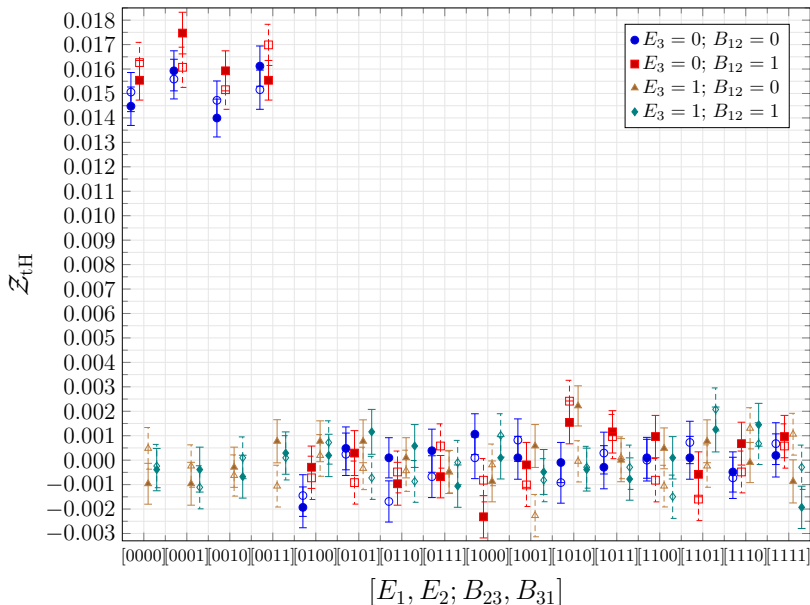
$$Z[B] \equiv \frac{N_{\text{conf with } B}}{N_{\text{total conf}}}$$

- ▶ Further, average over Euclidean orthogonal rotations if T^4 .
E.g., $(B_{12}, B_{13}, B_{14}, B_{23}, B_{24}, B_{34}) = (0, 0, 0, 1, 1, 1)$, then

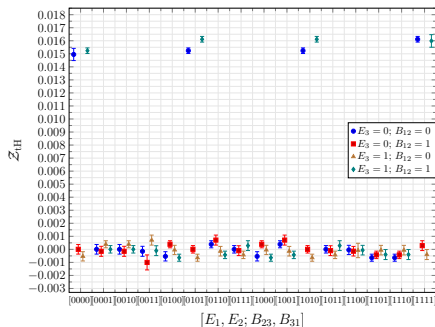
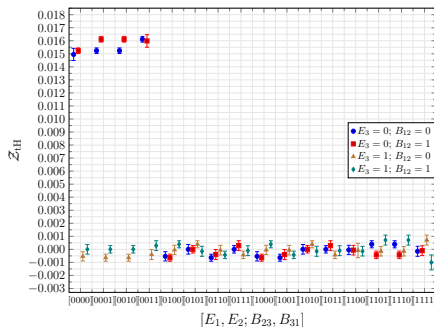
$$N_{\text{conf with } B} = \frac{1}{4}(N_{(0,0,0,1,1,1)} + N_{(0,1,1,0,0,1)} + N_{(1,0,1,0,1,0)} + N_{(1,1,0,1,0,0)}).$$

- We use 2590 configurations for $\beta = 2.6$ and $L = 20$
 - ▶ Note that $\mathcal{Z}_{\text{tH}} \geq 0$ if reflection positivity holds.

Confining phase in $\mathcal{Z}_{\text{tH}}[E, B]$ [OM–Suzuki '25]



Averaged $\mathcal{Z}_{\text{tH}}[E, B]$ and Witten effect



- Light or heavy flux

$$\mathcal{Z}_{\text{tH}}[\mathbf{E} = 0; \mathbf{B}] / \mathcal{Z}_{\text{tH}}[\mathbf{E} = 0; \mathbf{B} = 0] \sim 1$$

$$\mathcal{Z}_{\text{tH}}[\mathbf{E} \neq 0; \mathbf{B}] / \mathcal{Z}_{\text{tH}}[\mathbf{E} = 0; \mathbf{B} = 0] \sim 0$$

- Witten effect

$$\mathcal{Z}_{\text{tH}, \theta=2\pi}[E_i; \mathbf{B}] = \mathcal{Z}_{\text{tH}, \theta=0}[E_i + B_{jk}, \mathbf{B}]$$

Finite size scaling and finite temperature [in progress]

- String tension at large volume is given by

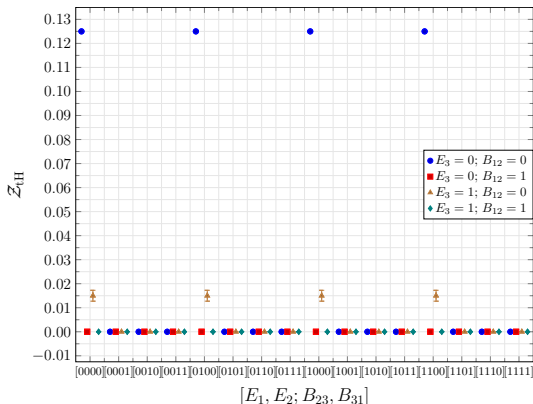
$$\mathcal{Z}_{\text{tH}}[\mathbf{E} = 0; \mathbf{B}] / \mathcal{Z}_{\text{tH}}[\mathbf{E} = 0; \mathbf{B} = 0] \sim 1 + O(e^{-\sigma L^2})$$

$$\mathcal{Z}_{\text{tH}}[\mathbf{E} \neq 0; \mathbf{B}] / \mathcal{Z}_{\text{tH}}[\mathbf{E} = 0; \mathbf{B} = 0] \sim O(e^{-\sigma L^2})$$

- ▶ More and more statistics needed
- When $T^3 \times S^1$ and near critical temperature T_c ?
 - ▶ Is $Z[B]$ truly partition function?
 - ▶ For fixed B , after enough thermalized, $Z[B] \rightarrow \mathcal{Z}[B]$.
 - ▶ “Separability” of $\sum_B \mathcal{Z}[B]$ is important.
 - ▶ Provide a good distribution of $P_F(B \rightarrow B')$!
 - ★ Let $U[B]$ be the thermalized configuration associated with B .
 - ★ Consider $H(B')$ with initial $U[B]$, measure thermalization time.
 - ★ $P_F(B \rightarrow B')$ should stay a B for enough MD time.

Finite temperature [preliminary of preliminary]

- Here, $B_{34} = z = 0, 1$; otherwise $B_{\mu\nu} = 0$
- Gaussian $P_F \sim e^{-\alpha(z-z')^2}$ with thermalization ~ 100 MD time
 - ▶ $\beta = 2.4$, box size $8^3 \times 4$



Summary

- Direct Monte Carlo simulation of $\mathcal{Z}_{\text{tH}}[\mathbf{E}; \mathbf{B}]$
 - ★ See [Nguyen–Tanizaki–Ünsal '23] for properties of $\mathcal{Z}_{\text{tH}}[\mathbf{E}; \mathbf{B}]$
 - ▶ (First) Attempt to measure all possible combinations
 - ▶ Light or heavy flux in confining phase
 - ▶ Oblique confinement by Witten effect
- For finite temperature, reproduce free energy near T_c
 - ▶ [Kovács–Tomboulis '00, de Forcrand–von Smekal '01]
 - ▶ Be careful to thermalization for each $\mathcal{Z}[B]$
- Related topics:
 - ▶ Numerical verification of Wilson–'t Hooft classification [Kanamori–OM–Nagai–Tanizaki–Tomiya–Watanabe wip]
 - ▶ Possible direct observations of partition function (continuum)
 - ★ Twisted p.f. from symmetry TFT [Maeda–Tanizaki '25]
- Quite simple. Feel free to play around with it!